

Design and Simulation of a Digital Access Card Verification System

Introduction

“Design and Simulation of a Digital Access Card Verification System” is a good Digital Logic / Digital Electronics case-study topic. It involves using logic gates, Boolean expressions, truth tables, and simulation to verify whether an access card should be accepted or rejected.

1. Basic Idea

A digital access card system checks certain conditions before allowing a person to enter a restricted area.

For example, suppose the system has these inputs:

Input	Meaning	Logic 1	Logic 0
C	Valid access card	Card is valid	Card is invalid
P	Correct PIN entered	Correct PIN	Incorrect PIN
A	Authorized user	User is authorized	User is not authorized

The **Access Granted (G)** output should become 1 only when all required conditions are satisfied.

$$G = C \cdot P \cdot A$$

Here, “.” means AND.

2. Truth Table

C	P	A	$G = C \cdot P \cdot A$	Result
0	0	0	0	Access Denied
0	0	1	0	Access Denied
0	1	0	0	Access Denied
0	1	1	0	Access Denied
1	0	0	0	Access Denied
1	0	1	0	Access Denied
1	1	0	0	Access Denied
1	1	1	1	Access Granted

3. Logic Circuit

The circuit can be designed using an AND gate:

Valid Card ■■■■

■

Correct PIN ■■■■■■ AND ■■■■ Access Granted

■

Authorized ■■■■

When $C = 1$, $P = 1$, and $A = 1$, the AND gate produces $G = 1$. Otherwise, $G = 0$.

4. Simulation

The circuit can be simulated using software such as **Logisim**, **Proteus**, **Multisim**, or similar digital-logic simulators.

During simulation, you can change the input switches:

$C = 1$

$P = 1$

$A = 1$

and observe that the **Access Granted LED turns ON**.

If any one input becomes 0, the output becomes 0 and the **Access Denied** indication can be activated.

5. Main Objectives of the Project

1. Designing a digital verification system.
2. Identifying inputs and outputs.
3. Creating a truth table.
4. Deriving a Boolean expression.
5. Implementing the circuit using logic gates.
6. Simulating and testing different input combinations.
7. Verifying that unauthorized access is rejected.